DEEP LEARNING CIE SUBMISSION

PBL TOPIC: ATTENTION-BASED TEXT CLASSIFICATION

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Project Topic:	Attention-Based Text Classification
Paper Reviewed:	*Attention Is All You Need*
Authors & Year:	Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, et al. (2017)
Publication Venue:	NeurIPS 2017 | arXiv:1706.03762

1. Problem Addressed

Sequential recurrence in RNNs/LSTMs prevents parallel training across GPU clusters and limits long-range context capture.

2. Motivation & Need

Recurrent timesteps force O(N) sequential operations. Replacing recurrence with self-attention enables O(1) parallel operations.

3. Methodology & Model Architecture

Introduced the Transformer architecture built entirely on Multi-Head Scaled Dot-Product Self-Attention and Positional Encoding.

4. Dataset & Experimental Setup

WMT 2014 English-to-German (4.5M pairs) and English-to-French (36M pairs).

5. Attention Mechanism Formulation

Scaled Dot-Product Formula: Attention(Q, K, V) = softmax((Q * K^T) / sqrt(d_k)) * V. Multi-Head Attention projects Q, K, V into h sub-spaces.

6. Key Results Reported in Paper

Achieved state-of-the-art 28.4 BLEU score on WMT En-De, training 3.5x faster than previous best models.

7. System Advantages & Strengths

Enables full parallel matrix execution, eliminates recurrent bottlenecks, and captures arbitrary distance dependencies.

8. Limitations & Constraints

Quadratic memory complexity O(N^2) with respect to input sequence length N.

9. Identified Research Gap

Recurrent neural architectures restricted parallelization during deep neural network training.

10. Relevance to Our PBL Project

Provides the mathematical basis for our interactive Q-K-V self-attention matrix demonstration module.

11. Student's Critical Observations

Transformer self-attention revolutionized deep learning, forming the foundation for BERT, GPT, and modern Large Language Models.

12. References

[1] A. Vaswani et al., 'Attention is all you need,' in NeurIPS, 2017.